

Solutions to MP363 Problem Set 2

(1)

P.1

For this problem, we want to show that each of the three inner products actually satisfies the necessary requirements. They obviously do, but it's always good to double-check...

(a) If $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ and $w = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}$, $v+w$ is $\begin{pmatrix} v_1+w_1 \\ \vdots \\ v_n+w_n \end{pmatrix}$, so the i th component is $(v+w)_i = v_i + w_i$. Thus,

$$(i) \langle u | v+w \rangle = \sum_i u_i^* (v_i + w_i) = \sum_i u_i^* v_i + \sum_i u_i^* w_i = \langle u | v \rangle + \langle u | w \rangle.$$

The same idea applies for $\langle u+v | w \rangle$:

$$(ii) \langle u+v | w \rangle = \sum_i (u_i + v_i)^* w_i = \sum_i u_i^* w_i + \sum_i v_i^* w_i = \langle u | w \rangle + \langle v | w \rangle.$$

If λ is a constant, then $\lambda v = \begin{pmatrix} \lambda v_1 \\ \vdots \\ \lambda v_n \end{pmatrix}$, so $(\lambda v)_i = \lambda v_i$. Thus,

$$(iii) \langle u | \lambda v \rangle = \sum_i u_i^* (\lambda v_i) = \lambda \sum_i u_i^* v_i = \lambda \langle u | v \rangle.$$

Similarly,

$$(iv) \langle \lambda u | v \rangle = \sum_i (\lambda u_i)^* v_i = \sum_i \lambda^* u_i^* v_i = \lambda^* \langle u | v \rangle.$$

Finally, if we take complex-conjugate $\langle u | v \rangle$, we see that

$$(v) \langle u | v \rangle^* = \left(\sum_i u_i^* v_i \right)^* = \sum_i (u_i^* v_i)^* = \sum_i u_i v_i^* = \sum_i v_i^* u_i = \langle v | u \rangle$$

and so $\langle u | v \rangle = \sum_i u_i^* v_i = u^\dagger \cdot v$ is indeed an inner product

(b) If two functions are added the result is just the sum of them

so: $(u+w)(x) = u(x) + w(x)$. Thus,

$$(i) \langle u | v+w \rangle = \int_{-\infty}^{\infty} u^*(x) (v(x) + w(x)) dx = \int_{-\infty}^{\infty} u^*(x) v(x) dx + \int_{-\infty}^{\infty} u^*(x) w(x) dx = \langle u | v \rangle + \langle u | w \rangle$$

and

$$(ii) \langle u+v | w \rangle = \int_{-\infty}^{\infty} (u(x) + v(x))^* w(x) dx = \int_{-\infty}^{\infty} (u^*(x) + v^*(x)) w(x) dx = \int_{-\infty}^{\infty} u^*(x) w(x) dx + \int_{-\infty}^{\infty} v^*(x) w(x) dx = \langle u | w \rangle + \langle v | w \rangle.$$

$(\lambda v)(x)$ is the function $\lambda v(x)$, so

$$(iii) \langle u | \lambda v \rangle = \int_{-\infty}^{\infty} u^*(x) (\lambda v(x)) dx = \lambda \int_{-\infty}^{\infty} u^*(x) v(x) dx = \lambda \langle u | v \rangle$$

and

$$(iv) \langle \lambda u | v \rangle = \int_{-\infty}^{\infty} (\lambda u(x))^* v(x) dx = \int_{-\infty}^{\infty} \lambda^* u^*(x) v(x) dx = \lambda^* \langle u | v \rangle.$$

Finally,

$$\langle u | v \rangle^* = \left(\int_{-\infty}^{\infty} u^*(x) v(x) dx \right)^* = \int_{-\infty}^{\infty} (u^*(x) v(x))^* dx = \int_{-\infty}^{\infty} u(x) v^*(x) dx = \langle v | u \rangle$$

②

$$= \int_{-\infty}^{\infty} u(x) v^*(x) dx = \int_{-\infty}^{\infty} v^*(x) u(x) dx = \langle u | v \rangle$$

because x is a real variable ($x^* = x$).

Thus, both of the expts define inner products which is good.

P2: There are a number of ways to prove all these properties of the adjoint operation, but here I'll do it by looking at the individual elements in the matrix. In other words, I'll use $(A^\dagger)_{ij} = A_{ji}^*$ rather than just $A^\dagger$. This means that the definition for adjoint gives

$$(A^\dagger)_{ij} = (A_{ji})^*$$

so we go

$$(a) (A+B)_{ij} = A_{ij} + B_{ij}, \text{ so } [(A+B)^\dagger]_{ij} = (A+B)_{ji}^*$$

$$= (A_{ji} + B_{ji})^* = (A_{ji})^* + (B_{ji})^* = (A^\dagger)_{ij} + (B^\dagger)_{ij} = (A^\dagger + B^\dagger)_{ij}$$

$$\text{and so } \boxed{(A+B)^\dagger = A^\dagger + B^\dagger}$$

$$(b) \alpha A \text{ is linear with elements } (\alpha A)_{ij} = \alpha A_{ij}, \text{ so}$$

$$(\alpha A^\dagger)_{ij} = (\alpha A)_{ji}^* = (\alpha A_{ji})^* = \alpha^* A_{ji}^* = \alpha^* (A^\dagger)_{ij}$$

$$= (\alpha^* A^\dagger)_{ij}$$

$$\text{and so } \boxed{(\alpha A)^\dagger = \alpha^* A^\dagger}$$

$$(c) (AB)_{ij} = \sum_k A_{ik} B_{kj}, \text{ because that's the definition of matrix multiplication. Thus,}$$

$$[(AB)^\dagger]_{ij} = [(AB)_{ji}]^* = \left[\sum_k A_{jk} B_{ki} \right]^*$$

$$= \sum_k (A_{jk})^* (B_{ki})^* = \sum_k (A^\dagger)_{kj} (B^\dagger)_{ik}$$

$$= \sum_k (B^\dagger)_{ik} (A^\dagger)_{kj} = (B^\dagger A^\dagger)_{ij}$$

$$\text{so } \boxed{(AB)^\dagger = B^\dagger A^\dagger}$$

(d) Finally, the fact that two adjoint operations return us to the original matrix:

$$\begin{aligned} ((A^\dagger)^\dagger)_{ij} &= ((A^\dagger)_{ji})^* = ((A_{ij})^*)^* = A_{ij} \\ \text{so } \boxed{(A^\dagger)^\dagger &= A} \end{aligned}$$

As stated, these hold for any of the operators we've talked about, not just matrices. So regardless of if A is a matrix, a differential operator, or whatever, $(\alpha A)^\dagger = \alpha^* A^\dagger$ and so on. Thus, for ex. $(\hat{X}P)^\dagger = P^\dagger \hat{X}^\dagger = P\hat{X}$ even when $\hat{X}$ & P are acting on functions rather than column vectors. Handy stuff to know.

P.3

Recall that the adjoint operation on matrices is the transpose-conjugate (or conjugate-transpose), and that the trace of a matrix is the sum of its diagonal entries. That's really all we need for this problem.

(a) $\langle M|N \rangle = \frac{1}{2} \text{tr}(M^\dagger N)$, so first we need the product:

$$\begin{aligned} M^\dagger N &= \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}^\dagger \begin{pmatrix} 2+i & 1 \\ -1 & i \end{pmatrix} = \left[\begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}^T \right]^* \begin{pmatrix} 2+i & 1 \\ -1 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}^* \begin{pmatrix} 2+i & 1 \\ -1 & i \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ -i & 1 \end{pmatrix} \begin{pmatrix} 2+i & 1 \\ -1 & i \end{pmatrix} = \begin{pmatrix} 2+i & 1 \\ -2i & 0 \end{pmatrix} \end{aligned}$$

$$\text{so } \langle M|N \rangle = \frac{1}{2} \text{tr} \begin{pmatrix} 2+i & 1 \\ -2i & 0 \end{pmatrix} = \frac{1}{2} (2+i) = \boxed{1+i/2}$$

Similarly,

$$\begin{aligned} N^\dagger M &= \begin{pmatrix} 2+i & 1 \\ -1 & i \end{pmatrix}^\dagger \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} = \left[\begin{pmatrix} 2+i & 1 \\ -1 & i \end{pmatrix}^T \right]^* \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2+i & -1 \\ 1 & i \end{pmatrix}^* \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2-i & -1 \\ 1 & -i \end{pmatrix} \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2-i & 2i \\ 1 & 0 \end{pmatrix} \end{aligned}$$

$$\text{so } \langle N|M \rangle = \frac{1}{2} \text{tr} \begin{pmatrix} 2-i & 2i \\ 1 & 0 \end{pmatrix} = \frac{1}{2} (2-i) = \boxed{1-i/2}$$

and since $(1+i/2)^* = 1-i/2$, we conclude that $\boxed{\langle M|N \rangle^* = \langle N|M \rangle}$

(b) The first part is easy:

$$\sigma_1^\dagger = \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^T \right]^* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_1$$

$$\sigma_2^\dagger = \left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}^T \right]^* = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}^* = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \sigma_2$$

$$\sigma_3^\dagger = \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^T \right]^* = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^* = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_3$$

so $\boxed{\text{all three are self-adjoint}}$

(4)

Since $\sigma_i^\dagger = \sigma_i$, then $\langle \sigma_i | \sigma_j \rangle = \frac{1}{2} \text{tr}(\sigma_i^\dagger \sigma_j) = \frac{1}{2} \text{tr}(\sigma_i \sigma_j)$, which makes the work easier. We need to check only $\langle \sigma_1 | \sigma_1 \rangle = \langle \sigma_2 | \sigma_2 \rangle = \langle \sigma_3 | \sigma_3 \rangle = 1$ and $\langle \sigma_1 | \sigma_2 \rangle = \langle \sigma_1 | \sigma_3 \rangle = \langle \sigma_2 | \sigma_3 \rangle = 0$, so let's do it:

$$\langle \sigma_1 | \sigma_1 \rangle = \frac{1}{2} \text{tr}(\sigma_1^2) = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = \frac{1}{2} \text{tr} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} \cdot 2 = 1 \quad \checkmark$$

$$\langle \sigma_2 | \sigma_2 \rangle = \frac{1}{2} \text{tr}(\sigma_2^2) = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right] = \frac{1}{2} \text{tr} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} \cdot 2 = 1 \quad \checkmark$$

$$\langle \sigma_3 | \sigma_3 \rangle = \frac{1}{2} \text{tr}(\sigma_3^2) = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = \frac{1}{2} \text{tr} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} \cdot 2 = 1 \quad \checkmark$$

$$\langle \sigma_1 | \sigma_2 \rangle = \frac{1}{2} \text{tr}(\sigma_1 \sigma_2) = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right] = \frac{1}{2} \text{tr} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = \frac{1}{2} \cdot 0 = 0 \quad \checkmark$$

$$\langle \sigma_1 | \sigma_3 \rangle = \frac{1}{2} \text{tr}(\sigma_1 \sigma_3) = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = \frac{1}{2} \text{tr} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \frac{1}{2} \cdot 0 = 0 \quad \checkmark$$

$$\langle \sigma_2 | \sigma_3 \rangle = \frac{1}{2} \text{tr}(\sigma_2 \sigma_3) = \frac{1}{2} \text{tr} \left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = \frac{1}{2} \text{tr} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = \frac{1}{2} \cdot 0 = 0 \quad \checkmark$$

Technically, we should check $\langle \sigma_2 | \sigma_1 \rangle = \langle \sigma_3 | \sigma_2 \rangle = \langle \sigma_3 | \sigma_1 \rangle = 0$ as well, but we've told that $\frac{1}{2} \text{tr}(M^\dagger N)$ is an inner product, i.e., it satisfies all the properties we looked at in Problem 1.

Notice, $\langle \sigma_i | \sigma_j \rangle = \langle \sigma_j | \sigma_i \rangle^*$, so if one of these is zero, the other is as well. Period.

$$\langle \sigma_2 | \sigma_1 \rangle = \langle \sigma_1 | \sigma_2 \rangle^* = 0^* = 0 \quad \checkmark$$

$$\langle \sigma_3 | \sigma_2 \rangle = \langle \sigma_2 | \sigma_3 \rangle^* = 0^* = 0 \quad \checkmark$$

$$\langle \sigma_3 | \sigma_1 \rangle = \langle \sigma_1 | \sigma_3 \rangle^* = 0^* = 0 \quad \checkmark$$

And so $\langle \sigma_i | \sigma_j \rangle = \delta_{ij}$ and the matrices are orthonormal.

P.4

We already know that we'll get $\langle \psi | H \psi \rangle = \langle \psi | H \psi \rangle$, hence $H^\dagger = H$ was already shown. But this verification of the proof is useful because it gives the general method of showing self-adjointness (or lack thereof) of differential operators, of which there are lots. So let's do it.

(a) This is easy, since it follows immediately from the product rule for derivatives:

$$\frac{\partial}{\partial x} \left[f \frac{\partial g}{\partial x} - \frac{\partial f}{\partial x} g \right] = \frac{\partial f}{\partial x} \frac{\partial g}{\partial x} + f \frac{\partial^2 g}{\partial x^2} - \frac{\partial^2 f}{\partial x^2} g - \frac{\partial f}{\partial x} \frac{\partial g}{\partial x}$$

$$= \left[f \frac{\partial^2 g}{\partial x^2} - \frac{\partial^2 f}{\partial x^2} g \right]$$

(5)

Prove it, manually.

(b) Now to use it: the two inner products are

$$\begin{aligned} \langle \psi | H \psi \rangle &= \int_{-\infty}^{\infty} \psi^* \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V \psi \right) dx \\ &= -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \psi^* \frac{\partial^2 \psi}{\partial x^2} dx + \int_{-\infty}^{\infty} V \psi^* \psi dx \end{aligned}$$

and

$$\begin{aligned} \langle H \psi | \psi \rangle &= \int_{-\infty}^{\infty} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + V \psi^* \right) \psi dx \\ &= \int_{-\infty}^{\infty} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + V \psi^* \right) \psi dx \\ &= -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \frac{\partial^2 \psi^*}{\partial x^2} \psi dx + \int_{-\infty}^{\infty} V \psi^* \psi dx \end{aligned}$$

because V is real, as stated. Note that the second term in each is the same, and so will cancel out when we subtract the second from the first. Thus,

we have

$$\begin{aligned} \langle \psi | H \psi \rangle - \langle H \psi | \psi \rangle &= -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \psi^* \frac{\partial^2 \psi}{\partial x^2} dx + \frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \frac{\partial^2 \psi^*}{\partial x^2} \psi dx \\ &= -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \left(\psi^* \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi^*}{\partial x^2} \psi \right) dx \end{aligned}$$

But using the result from (a), $\psi^* \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi^*}{\partial x^2} \psi = \frac{\partial}{\partial x} \left(\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right)$. Thus,

$$\begin{aligned} \langle \psi | H \psi \rangle - \langle H \psi | \psi \rangle &= -\frac{\hbar^2}{2m} \int_{-\infty}^{\infty} \frac{\partial}{\partial x} \left(\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right) dx \\ &= -\frac{\hbar^2}{2m} \left[\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right]_{-\infty}^{\infty} \end{aligned}$$

from the Fundamental Theorem of Calculus: $\int_a^b \frac{df}{dx}(x) dx = [f(x)]_a^b = f(b) - f(a)$.

Now recall that all states must be normalized, i.e. $\|\psi\| = \sqrt{\langle \psi | \psi \rangle} = 1$. For this integral to exist, all wavefunctions must vanish as x goes to either $+\infty$ or $-\infty$. Thus, $\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \rightarrow 0$ both as $x \rightarrow \infty$ and $x \rightarrow -\infty$, so our result vanishes. The difference between the inner products is zero, and so $\langle \psi | H \psi \rangle = \langle H \psi | \psi \rangle$ for all ψ and ψ . Therefore, $[H \text{ is self-adjoint}]$.